

Specialization:

Network Design & Security

Business Focus:Secure Network Infrastructure for Small
Businesses**Skill Outcome:**

- Network design and basic LAN setup
- IP addressing and device configuration
- Connectivity testing and troubleshooting

DATA BRIDGE INNOVATIONS**Building a SOHO Network: Introduction to
Networking and Network Security****Project Learning Opportunities:**

- Learn how to design and configure a small routed network.
- Understand basic IP addressing and inter-network communication.
- Gain hands-on experience using Cisco Packet Tracer for network simulation.

Tools and Technology Used:

Company Introduction

You have been hired as a junior **network technician** to design a small office network for a company called DataBridge Innovations.

The office has three departments:

- **Admin Department** – handles internal operations
- **Sales Department** – handles customer communication
- **HR Department** – handles recruitment tasks

All departments must be connected so that devices can communicate across the network.

The company wants the network designed and tested using **Cisco Packet Tracer**, ensuring that:

- All devices are correctly connected
- IP addresses are properly assigned
- Basic connectivity is confirmed using simple network tests



The Business Problem

DataBridge Innovations is expanding its operations and requires a reliable internal network to support communication across its three key departments: Admin, Sales, and HR. Currently, there is no structured network infrastructure, which limits collaboration, slows down operations, and increases the risk of miscommunication and inefficiency.

The Problem

The company lacks a properly designed Local Area Network (LAN) that enables seamless communication between departments. Without this:

- Devices cannot communicate effectively across departments
- There is no proper IP addressing scheme
- Network performance and reliability cannot be guaranteed
- Troubleshooting and scalability become difficult

Your Role

As a **Junior Network Technician**, your responsibility is to:

- Design a functional network topology using Cisco Packet Tracer
- Connect all departmental devices appropriately (PCs, switches, routers if needed)
- Assign correct IP addresses to all devices
- Ensure inter-department communication is possible
- Test and verify connectivity using basic tools like ping



Rationale for the Project

This Project should....

Rationale 1



Improve Communication Efficiency

A structured network ensures that Admin, Sales, and HR can share information quickly and reliably, improving overall workflow.

Rationale 2



Enhance Network Organization

Segmenting departments logically (even in a small network) allows better management, easier troubleshooting, and future scalability.

Rationale 3



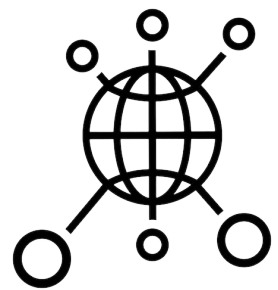
Support Business Growth

A properly designed network provides a foundation for future expansion, such as adding more users, devices, or services.



Project Initiative

By completing this project, you will be able to:



Network Design & Topology Creation

- Create a simple LAN topology using switches to connect all departments
- Ensure proper physical and logical layout



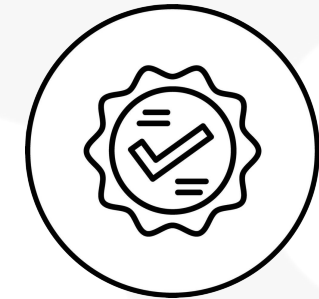
Device Configuration

- Assign IP addresses to all devices using a consistent addressing scheme
- Configure basic settings on devices



Connectivity Implementation

- Ensure all devices are properly connected using appropriate cables
- Enable communication between departments



Testing & Validation

- Use ping tests to confirm connectivity
- Troubleshoot any communication failures
- Validate that all departments can communicate successfully

Network Overview

Department	Network/Subnet	Devices
Admin	192.168.10.0/24	2 PCs & 1 server
Sales	192.168.20.0/24	2 PCs
HR	192.168.30.0/24	2 PCs

Router(2911): Connects all three networks

Switches(2960): One per department

Project Workflow

Step 1: Network Setup

- Open Cisco Packet Tracer.
- Add 1 Router(2911), 3 Switches, and 6 PCs (2 per department).
- Connect the devices using straight-through cables.

Step 2: IP Address Configuration

- Assign IP addresses to each PC and corresponding router interfaces.

Example:

- Admin: 192.168.10.1 (router), 192.168.10.10–10.11 (PCs)
- Sales: 192.168.20.1 (router), 192.168.20.10–20.11 (PCs)
- HR: 192.168.30.1 (router), 192.168.30.10–30.11 (PCs)

Step 3: Test Connectivity

Use the ping command between departments.

- At this stage, all PCs should communicate successfully.

→ Expected Outcome



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At the end of this practical, participants will understand:

- How a basic routed network works
- How IP addressing enables communication
- How to test and troubleshoot simple network Connectivity

